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TRW Automotive

Slip Control Software Design

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REFERENCES

Ref.	Document	Title	Ver.	Location

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1 ABSTRACT

This module adapts [ARM](#) and [AyLimiter](#) parameters according to the vehicle configuration. The configuration is governed by current chassis level, vehicle load and YSC Brake Modes.

2 ACRONYMS

ARM Active Roll Mitigation

AyLimiter Control Algorithm that limit the Lateral Acceleration of the Vehcile

COG Center of Gravity

RSC Roll Stability Control

ESC Electronic Stability Control

YSC Yaw Stability Control

WSR Wheel Slip Regulator

3 OVERVIEW

This document describes the purpose and behavior of Arm_Adaptation.

3.1 DESIGN GOAL

[ARM](#) trims are calibrated in order to prevent a vehicle with worst case configuration (high [COG](#), chassis) under worst case conditions(tire, loading, surface) from roll over. A driver of a vehicle with very low roll over propensity (vehicle configuration i.e.: low [COG](#)) could notice [ARM](#) interventions to be uncomfortable, exaggerated and disturbing. In this module [ARM](#) and [AyLimiter](#) parameters are adjusted due to information about the vehicle configuration.

3.2 GENERAL REQUIREMENTS

1. [ARM](#) should prevent dynamic rollover.
2. [ARM](#) should prevent static rollover.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when [ARM](#) is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

None.

4 DETAILED DESIGN

This chapter outlines the detailed design of Arm_Adaptation.

4.1 ALGORITHM DESCRIPTION

This module is very specific to the customer, project or vehicle because it is dependant on the roll over propensity for all available vehicle configurations and the available information about different vehicle configurations.

In this core version the following input signals are supported:

- roof load information
- vehicle load information
- chassis level state = hight of [COG](#)

This information is used to adjust the [AyLimiter](#) intervention threshold.

5 PROVIDED TASKS

5.1 ArmAdaptation

Description	ArmAdaptation
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void ArmAdaptation (void)

6 PROVIDED SERVICES

No provided services.

7 CONFIGURATION OPTIONS

None.

8 CONFIGURATION PROCEDURES

None.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

None.

11 APPENDIX